LAB 01 ANSWERS REPORT

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1. **What are the key features of PyTorch?**

    PyTorch offers a wide range of features including

    - Dynamic computation graphs that are constructed during the forward pass, providing flexibility when developing and debugging neural networks.
    - Extensive library of pre-built neural network models and tools
    - Strong support for GPU and other hardware acceleration

2. **How do you create a tensor in PyTorch?**
    - You can create a tensor in Pytorch by using torch.sensor()

3. **What is the difference between element-wise multiplication and matrix multiplication in PyTorch?**
    - Element-wise multiplication multiples individual values using identical shapes
    - Matrix multiple computes the dot produces of rows and columns requiring aligned dimensions

4. **How can you perform tensor slicing and indexing in PyTorch?**
    - Tensor indexing and slicing can be done using the colon operator
    - This can include
        i. Single elements: tensor[row, col]
        ii. Range: tensor[start: end]
        iii. Entire columns: E.g. tensor[:,1]

5. **Why is it recommended to use a virtual environment when installing PyTorch?**
    - Ensure that a project's dependencies are self-contained, meaning that the packages required for one project do not conflict with those needed for another project on the same machine